

23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : 11

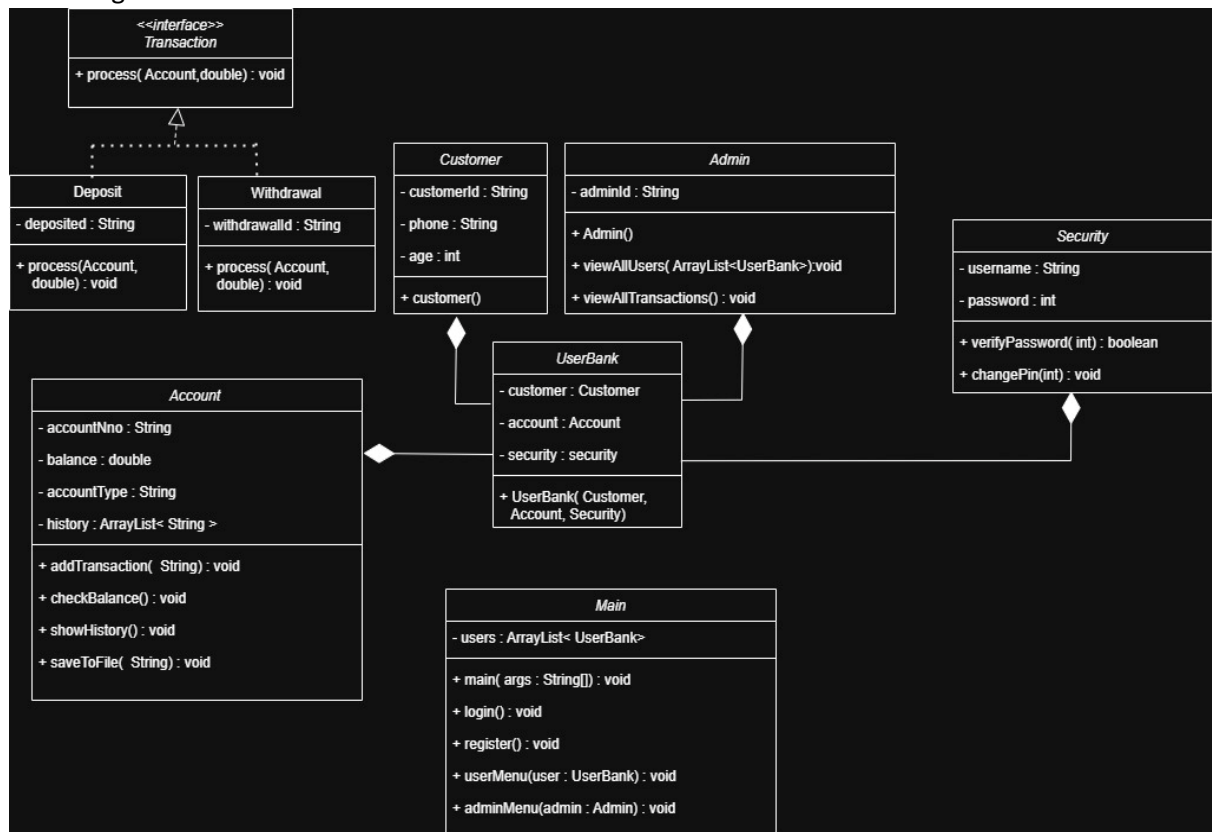
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1. Problem Title : INTERNET BANKING SYSTEM

STRUCTURAL DESIGN

2. Class diagram



3.

BEHAVIUIORAL DESIGN

4. Use-case diagram

a. List of actors

1. User

The user is a customer who performs banking operations such as checking balance, depositing money, withdrawing money, and viewing transaction history.

2. Admin

The admin manages user accounts and monitors transactions within the system.

b. List of use-cases

1. Register

Allows a new user to create an account in the internet banking system by providing required details.

2. Login

Enables the user or admin to log into the system using valid credentials.

3. Verify Login

Authenticates the user or admin credentials before granting access to the system.

4. Deposit

Allows the user to deposit money into their bank account.

5. Withdraw

Enables the user to withdraw money from their account.

6. Check Balance

Allows the user to view the current balance in their account.

7. View Transaction History

Displays all past transactions performed by the user.

8. Change PIN

Allows the user to securely change their account PIN.

9. Logout

Enables the user or admin to safely exit from the system.

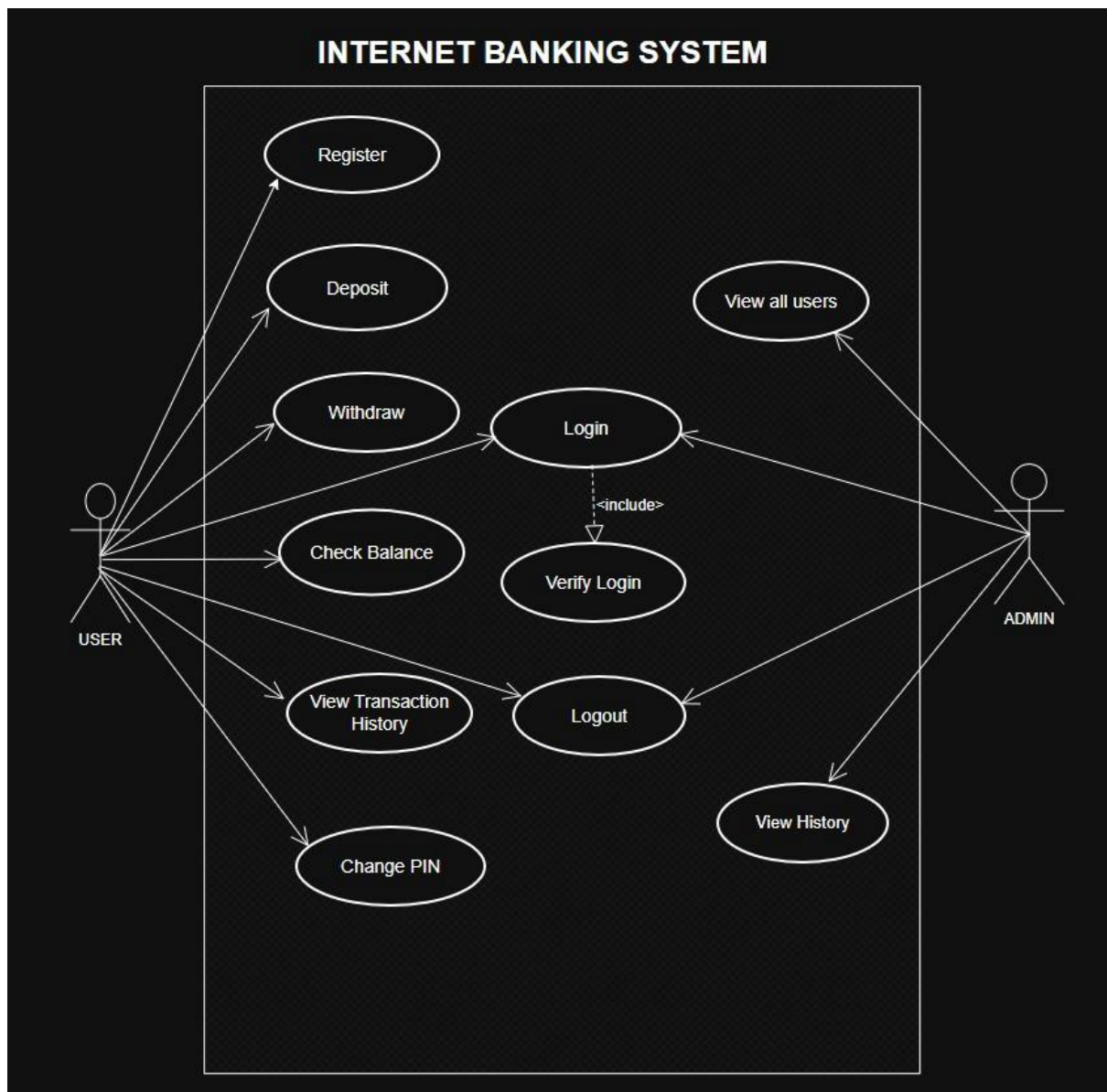
10. View All Users (Admin)

Allows the admin to view details of all registered users.

11. View History (Admin)

Enables the admin to monitor and view transaction history of users.

c. Use case diagram (use UML notations)

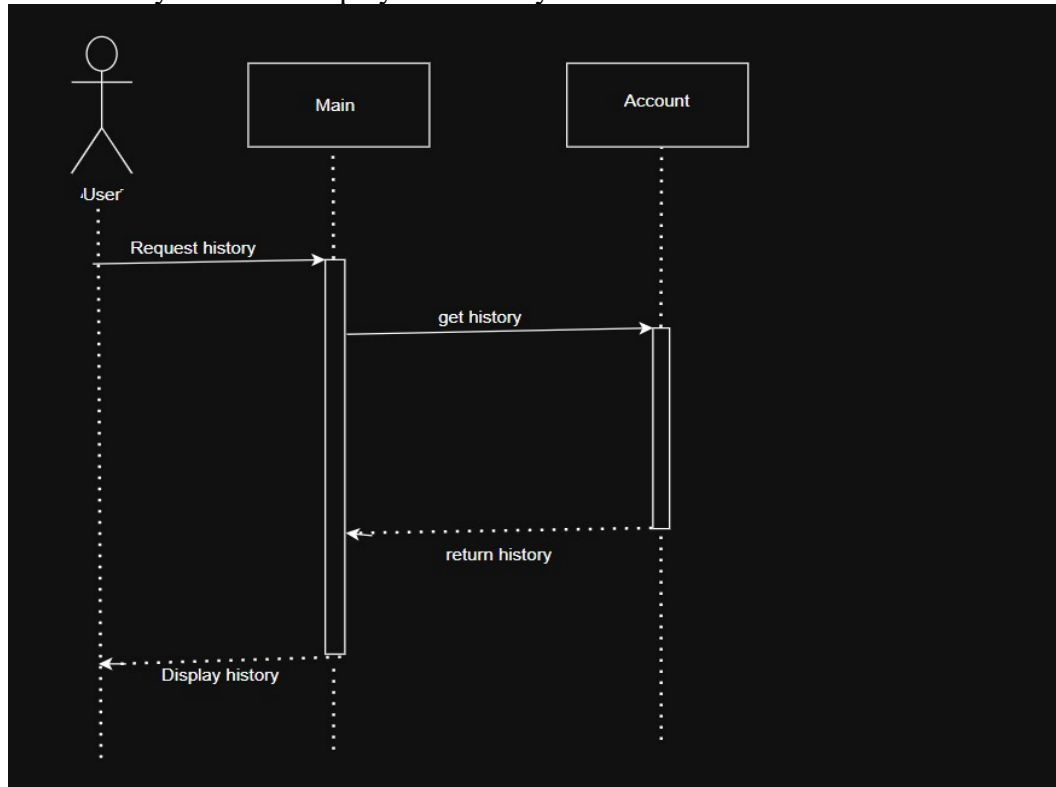


d. Sequence diagram (use UML notations)

a. Use case: View Transaction History

This diagram represents the process of viewing transaction history. The user requests the history from the main system, which forwards the request to the Account class. The Account retrieves the stored transaction history and sends it

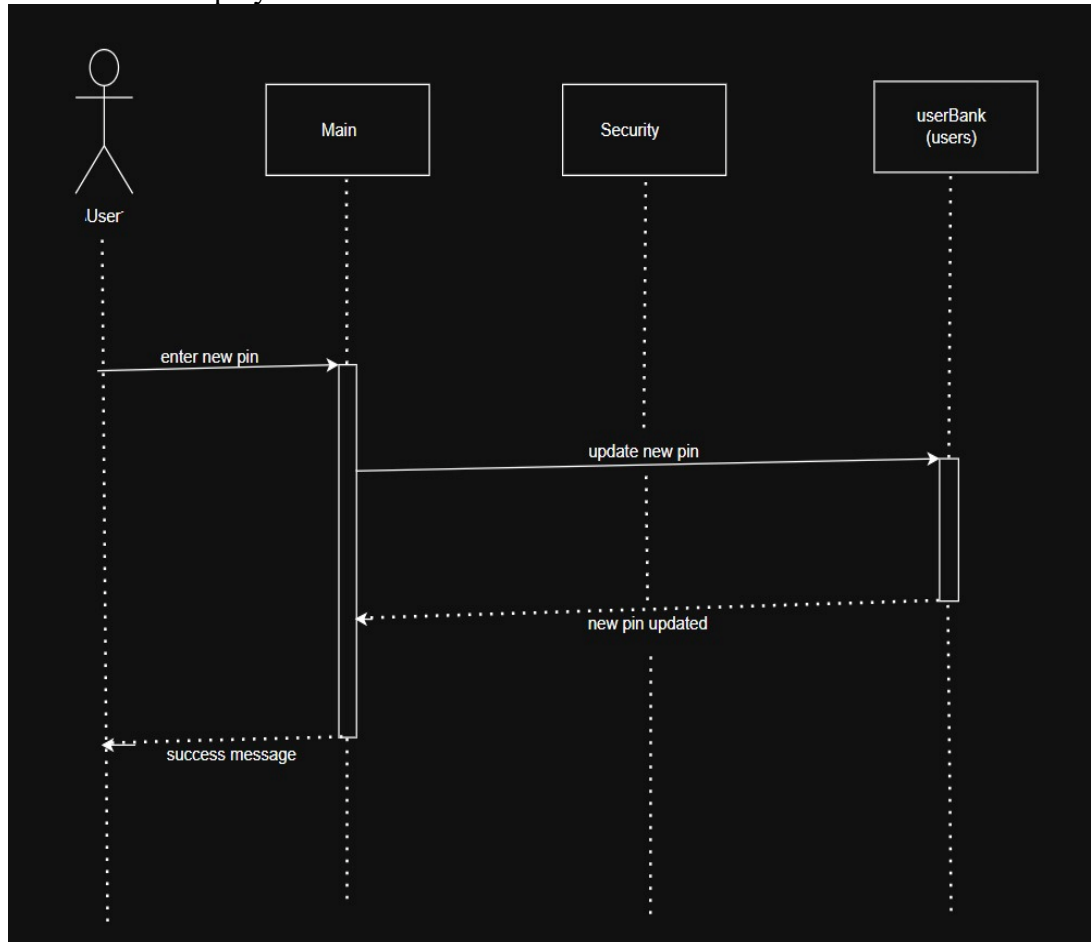
back. The system then displays the history to the user.



b. Use case: Change PIN

This diagram illustrates the process of changing the user's PIN. The user enters a new PIN through the system interface. The system sends the request to the Security class to update the PIN. Once updated, a confirmation message is

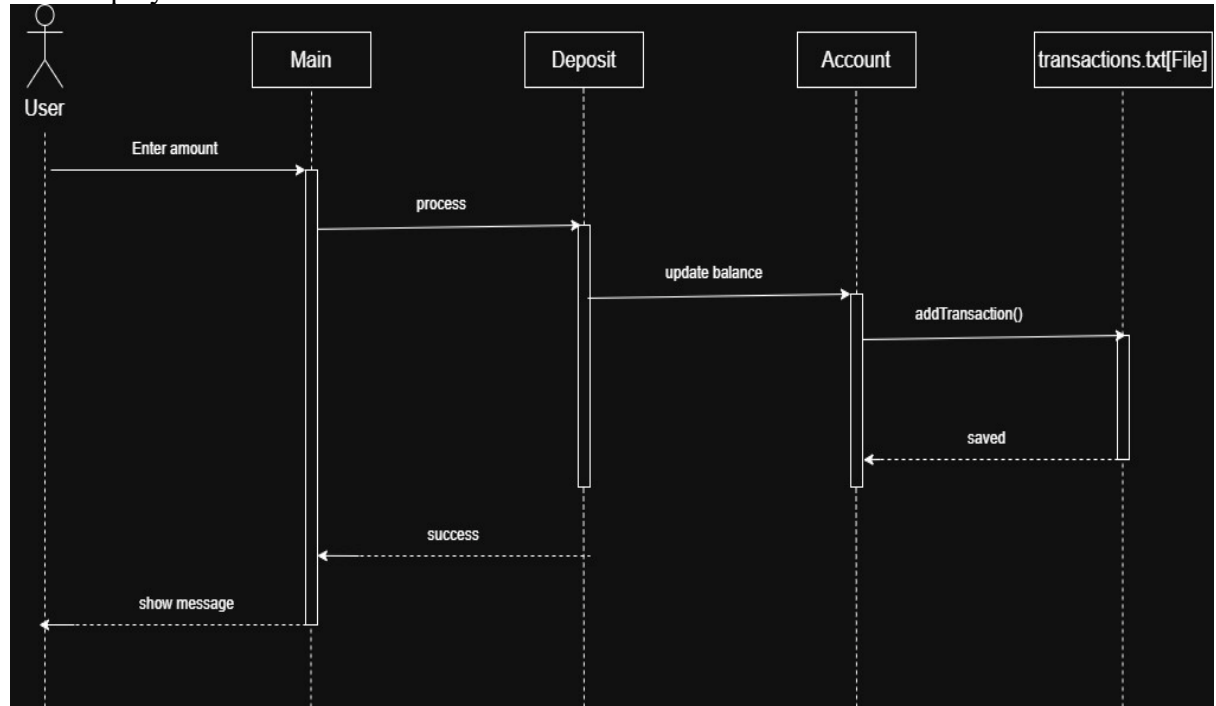
returned and displayed to the user.



c. Use case: Deposit Money

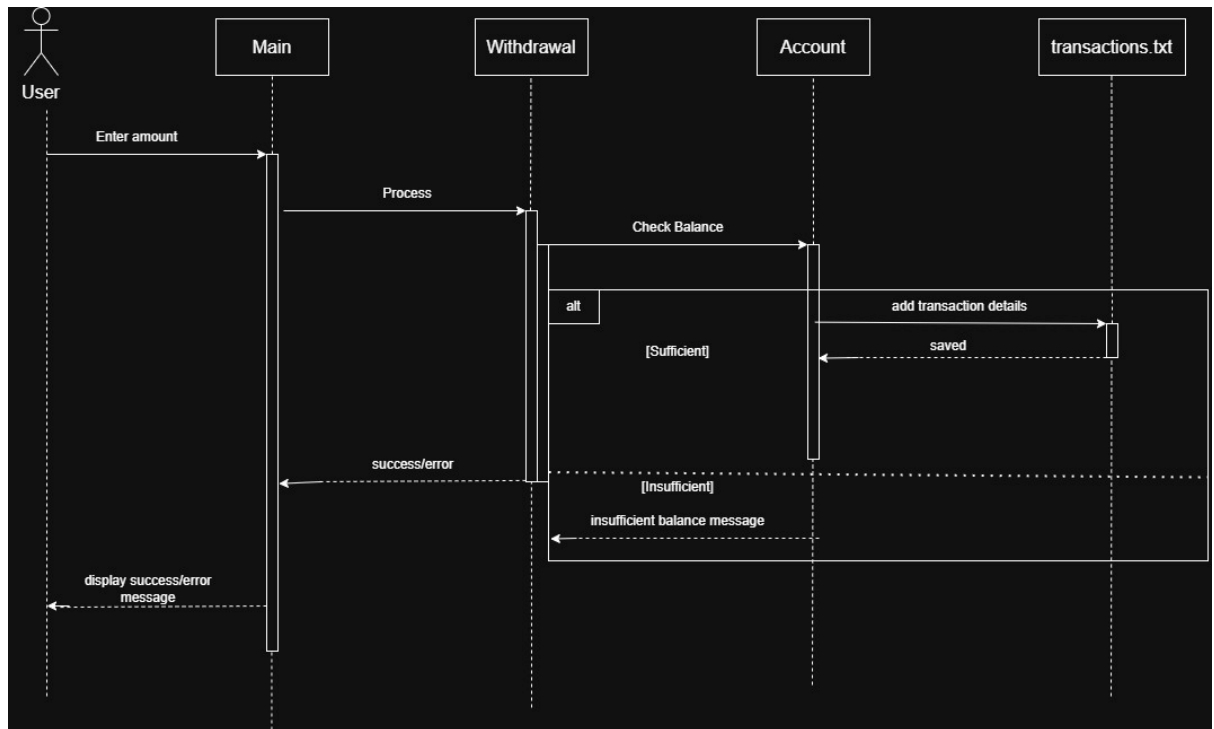
This diagram explains how money is deposited into the account. The user enters the amount, and the system calls the Deposit class to process it. The Account balance is updated, and the transaction is saved to a file. A success message is

then displayed to the user.



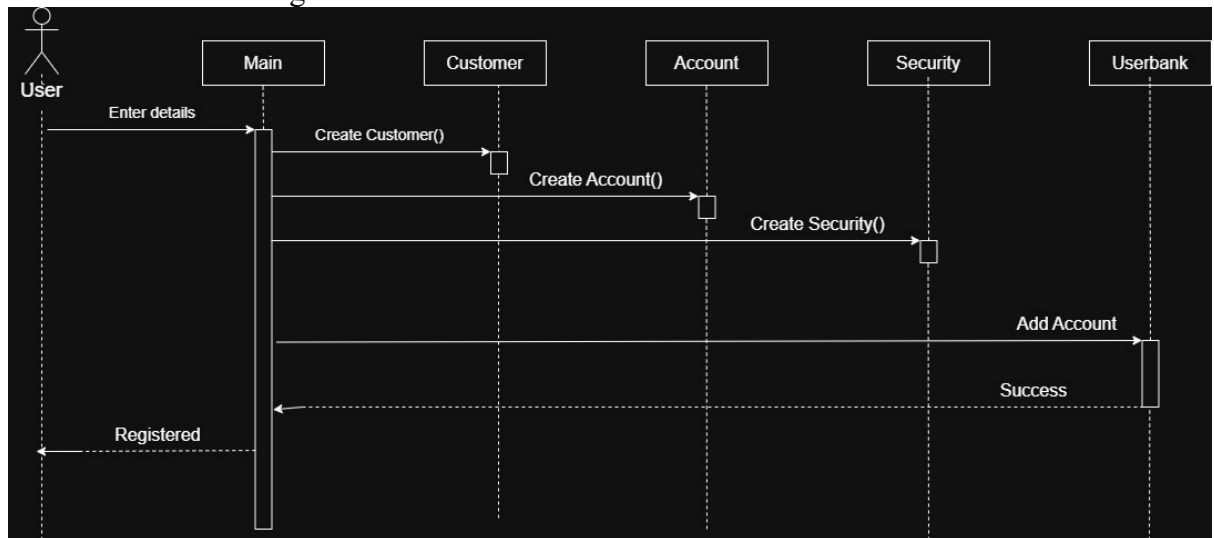
d. Use case: Withdraw Money

This diagram shows how money is withdrawn from the account. The user enters the withdrawal amount, and the system forwards the request to the Withdrawal class. The system checks the account balance, and if sufficient, updates the balance and stores the transaction in a file. If insufficient, an error message is shown; otherwise, success is displayed..



e. Use case: User Registration

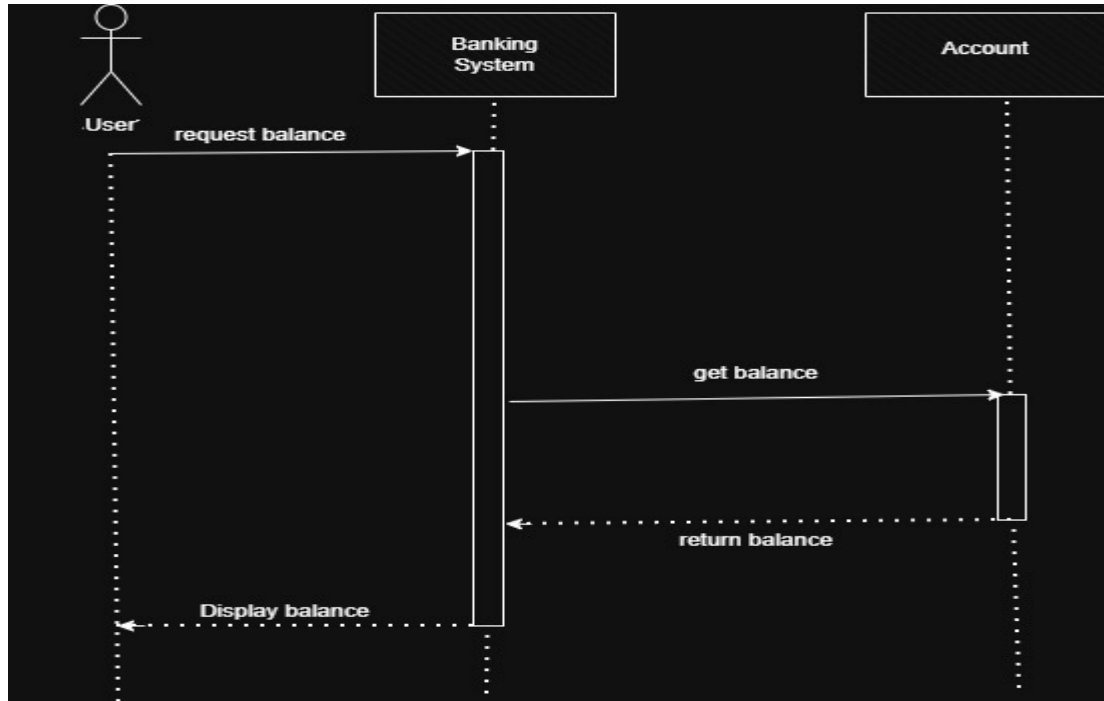
This diagram describes the user registration process. The user enters details, and the system creates Customer, Account, and Security objects. These are combined into a UserBank object and stored in the system. A success message is then returned to confirm registration.



f. Use case: Check Balance

This diagram shows how a user checks their account balance. The user sends a request to the class, which then calls the Account class to retrieve the balance. The Account returns the balance to the system. Finally, the system displays the balance

back to the user.



g. Use case: Login

This diagram represents the login process. The user enters username and PIN, and the system searches for the user in the UserBank list. The Security module verifies the PIN. If valid, access is granted and the menu is displayed; otherwise, an error message is shown.

